PREDICTING STOCK MARKET RETURNS

A PREPRINT

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ABSTRACT

We seek to investigate the question of stock market prediction by proposing an uncertainty-aware, volatility-constrained approach to forecasting daily excess returns of the S&P 500 and converting those forecasts into allocations of funds. The core idea is to model both expected excess return and its uncertainty, then trade selectively using feature-driven time-series models. We plan to utilize the Hull Tactical Market Prediction Kaggle competition as a framework, using the provided train and test sets for data. Evaluation will be determined by the R^2 value and correlation of predictions to actual forward market returns. The goal is to maximize the R^2 value while also finding a strong correlation between the actual and predicted returns.

1 Introduction

Traditionally, it has been understood that timing the market is irresponsible and foolhardy at best. However, this project aims to test whether modern machine learning models can time broad-market exposure responsibly. For each trading day t, we observe a multi-dimensional vector of market, macro, rates, price/valuation, volatility, sentiment, momentum, and dummy features (the dataset's $M*, E*, I*, P*, V*, S*, MOM*, D*$ families). We must choose an allocation $a_t \in [0, 2]$ to the S&P 500 to maximize a Sharpe-like objective while keeping realized portfolio volatility at or below 120% of the index's volatility. Labels are forward returns and forward excess returns (relative to a rolling 5-year mean), provided in train; a lagged version is available during evaluation to avoid look-ahead. The evaluation is strictly walk-forward via an API that serves features in chronological order.

2 Related Works

2.1 Asset Allocation Taking into Account the Time Lag of Volatility Forecasting (2011)

Friedl (2011) examines how delays between forecasting volatility and executing trades affect portfolio risk. The paper introduces a "Risk-at-Risk" (RaR) estimator—analogous to Value-at-Risk but applied to volatility—to account for forecast latency under a volatility cap. Using intraday data, the study shows that incorporating lag-adjusted volatility distributions leads to more stable, risk-constrained allocations. This work supports our project's emphasis on volatility caps and time-sensitive risk control, providing a theoretical basis for latency-aware allocation rules.

2.2 Momentum and Markowitz: A Golden Combination (2015)

Mean-Variance Optimization was a theory presented by Markowitz (1952), but considered impractical and unstable. However, this paper merges the Mean-Variance Optimization theory with momentum investing. It illustrates how returns can be improved by managing volatility and drawdown. The paper goes beyond classical theory and considers the practicality of investing. This paper is foundational literature for our project, giving some historical context on investments in risky assets under volatility constraints.

2.3 Adaptive learning for financial markets mixing model-based and model-free RL for volatility targeting (2021)

This paper explores model based and model-free reinforcement learning in a financial markets setting. The experiment in the paper aligns closely with our project requirements of only considering past data, and evaluates the model based on Sharpe ratios and maximum drawdown over volatility. The results of the paper show that a model-based implementation effectively captured volatility dynamics in the markets, and further, that a combined approach of model-free and model-based RL is most effective when considering volatility, macro, and risk appetite signals as contextual information. The researchers also conducted a features sensitivity analysis, which validated the importance of volatility and contextual variables in the performance of the model. The findings of this paper reinforce the value of dynamic, ML-driven (specifically RL-driven) volatility management when allocating between varying exposure levels, validating our confidence-weighted, volatility-aware framework.

2.4 When Uncertainty and Volatility Are Disconnected (2021)

Aït-Sahalia et al. (2021) distinguish between market volatility and broader economic or model uncertainty, showing that the two are not always aligned. Through empirical analysis, they demonstrate that treating volatility and uncertainty as distinct factors can significantly improve portfolio allocation outcomes. This study provides theoretical support for our inclusion of confidence scores alongside volatility forecasts, emphasizing that accounting for both sources of risk leads to more effective risk-adjusted decision-making.

2.5 Spatial Temporal Multivariate Mixture Density Networks for Volatility-Aware Portfolio Optimization (2025)

This paper proposes a deep learning framework (STMMDN) that combines LSTMs, graph neural networks, and mixture density outputs to forecast full multivariate return distributions. The model distinguishes normal versus volatile market regimes and integrates these predictions into allocation decisions. This paper aligns closely with our goal of an ML allocator that is aware of volatility by demonstrating how distributional forecasting and regime detection improve risk-adjusted portfolio performance.

3 Experimental/Theoretical Results

Briefly describe what method(s) your group used to assess your problem of interest and the empirical settings in which you evaluated your findings. Show plots, theorems, and/or tables of the performance of your method(s) and interpret what they mean; be sure to label all of your figures and explain the findings. You must also define any performance metrics you used for evaluation.

3.1 Implementation 1: Gradient Boosting

Correlation Coefficient: 0.7124
Cross-validation R-squared: -0.9520

One implemented method was a Gradient Boosting Regressor model. The model was trained on historical market data from the preprocessed train.csv dataset. Preprocessing steps included feature engineering (creating new features based on existing ones), handling missing values through exponential weighted average imputation, and standardizing features using StandardScaler.

To assess the model's performance during training, we utilize a 10-fold cross-validation. We partitioned training data into 10 subsets, iteratively training the model on 9 subsets and validating on the remaining 1, repeating this process 10 times to provide a more reliable estimate of performance. The final trained model's ability to generalize to unseen data was then evaluated on an independent, held-out test set (test.csv).

For predicting the market forward excess returns, we employed a Gradient Boosting Regressor (an ensemble learning method builds a strong predictive model from weaker models, sequentially adding predictors). The key parameters we configured for our model were:

- Number of Estimators: 1000 (number of boosting stages)
- Maximum Depth: 6 (depth of individual regression estimators)
- Learning Rate: 0.1 (shrinks contribution of each tree)
- Loss Function: 'squared_error' (optimizes for mean squared error)

To assess the performance of our Gradient Boosting Regressor, we primarily utilized the correlation coefficient to measure the linear correlation between the actual market forward excess returns and the model's predicted returns. A coefficient close to 1 indicates a strong positive linear relationship, meaning predictions move in the same direction as actual values, which is highly desirable for market prediction. We also use the R^2 cross validation error to understand how well our model captures and explains the variability across different periods.

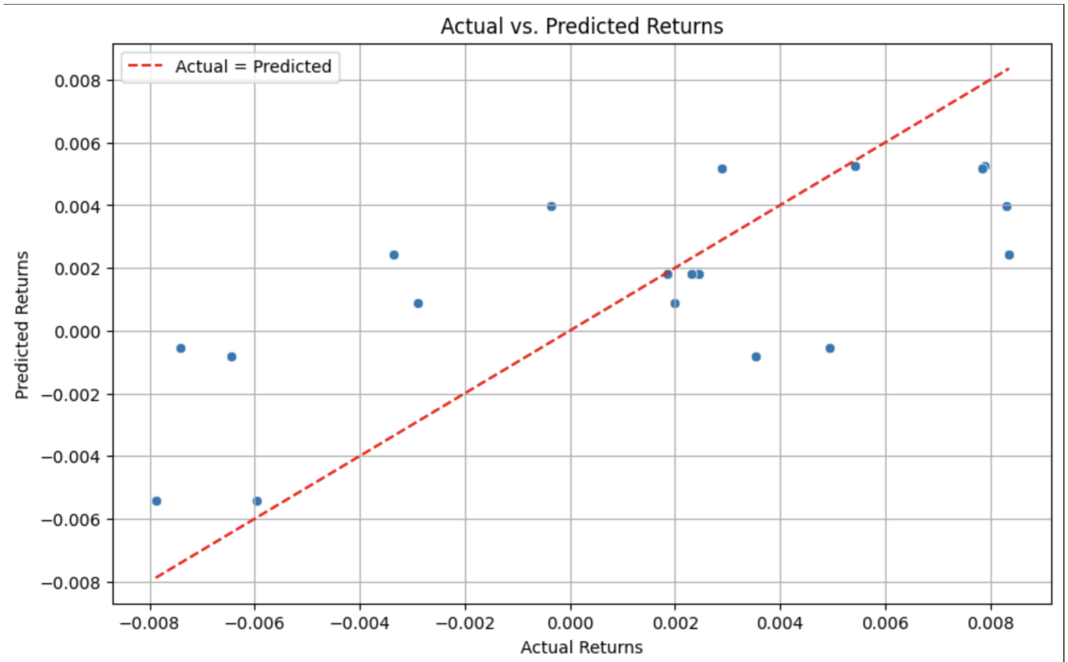

Figure 1: Illustrates the direct relationship between actual and predicted values in Gradient Boost

In our plot above, the points clustered along the red diagonal line (where Actual = Predicted) would signify perfect prediction. Our plot shows a general trend along this line, consistent with the positive correlation, but also demonstrates spread, indicating prediction errors.

The residual plot displays the prediction errors (residuals) against the date_id for the test set. Ideally, residuals are randomly scattered around zero with no discernible patterns. Any patterns, such as increasing or decreasing variance or a clear trend, would suggest potential biases or uncaptured relationships in the model. Our residual plot shows some significant scatter, though ideally we would want it to be more centered around zero.

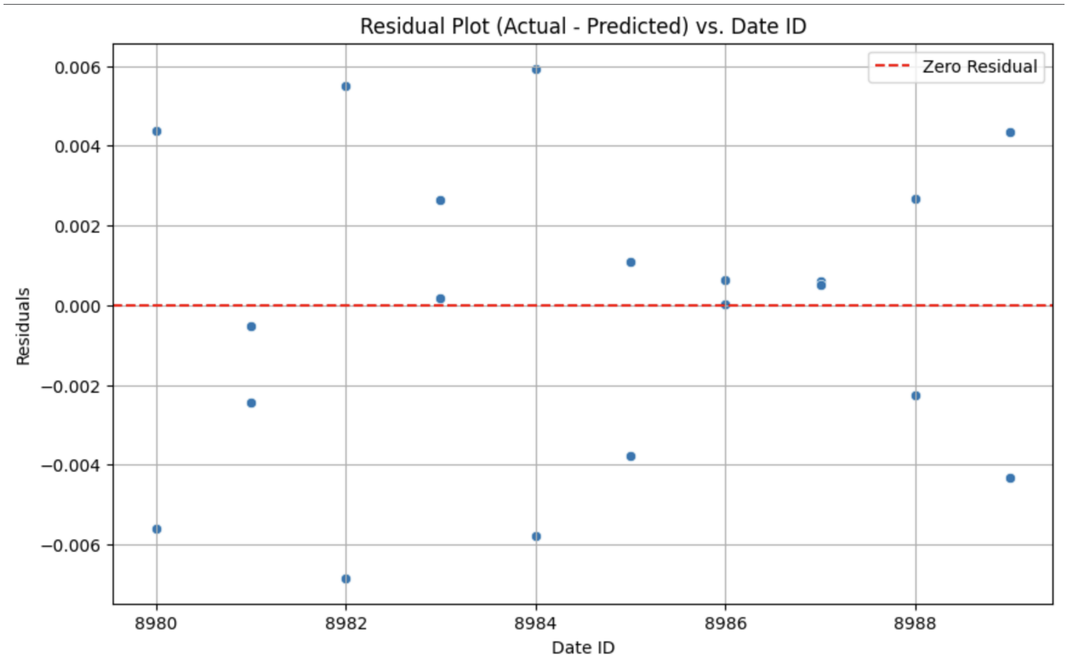

Figure 2: Residual plot for Gradient Boost

3.2 Implementation 2: CATBoost

Correlation Coefficient: 0.6732 Cross-validation R-Squared: -.5232

Another implementation method we attempted was a CATBoost model. We thought this model would be effective given the large amount of categorical data given to us in the training data set. The model was exclusively trained on the train.csv dataset given to us by the Kaggle competition. We used similar data prepossessing steps as we did in the Gradient Boosting model. Additionally, we used a similar 10-fold time series cross-validation R^2 metric to evaluate the model. Lastly, we used a CATBoost Regressor to predict the market forward excess returns. The main parameters that were configured for the model were:

- Number of iterations: 1000 (number of boosting stages)
- Maximum Depth: 6 (depth of individual regression estimators)
- Learning Rate: 0.1 (shrinks contribution of each tree)
- Loss Function: 'squared_error' (optimizes for mean squared error)

Consider the following graph between the actual returns and the predicted returns from our CATBoost model

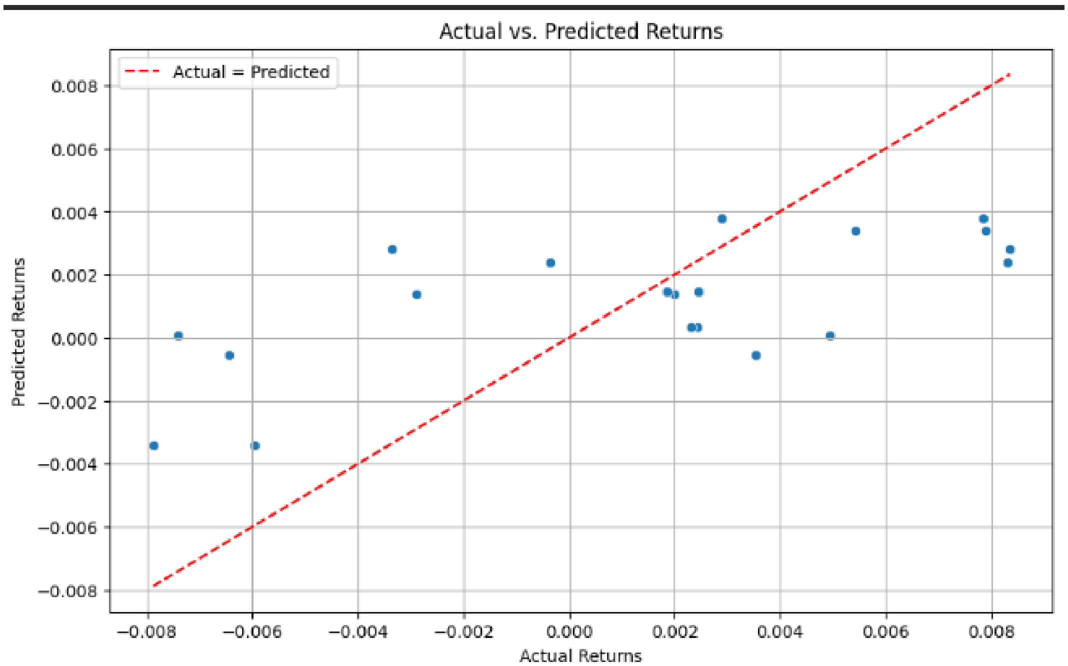

Figure 3: Illustrates the direct relationship between actual and predicted values

Additionally, consider the residual plot. This plot shows the difference between the CATBoosting model's estimtions and the actual data values.

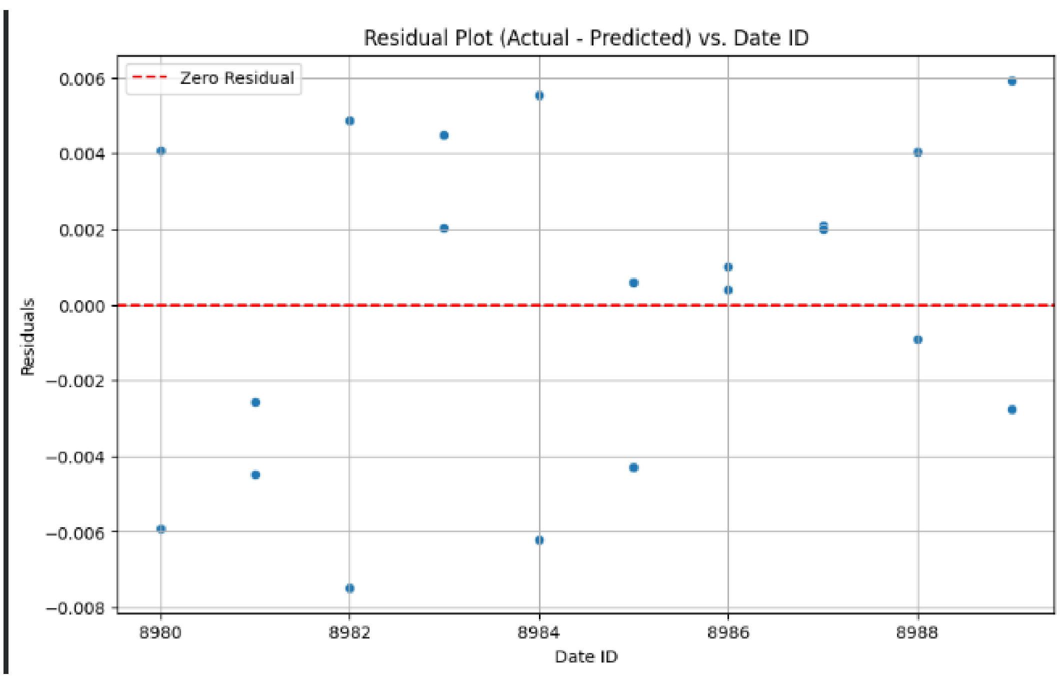

Figure 4: Residual plot

3.3 Implementation 3: XGBoost

Correlation Coefficient: 0.1209 Cross-validation R-Squared: -0.4165

The last implementation approach we implemented was using an XGBoost model. XGBoost models are commonly used over other modeling approaches due to their speed at large-scales and built-in regularization which helps prevent overfitting. The model was trained on the train.csv dataset provided by the Kaggle Hull Tactical Market Prediction competition. Preprocessing was performed similar to the previous implementations described. Similar to the previous implementations, we used a 10-fold time series cross-validation R^2 metric to evaluate the model. Lastly, we used an XGBoost regressor to predict market forward excess returns. The implementation's parameters were as follows:

- Number of iterations: 1000 (number of boosting stages)
- Maximum Depth: 6 (depth of individual regression estimators)
- Learning Rate: 0.05 (shrinks contribution of each tree)
- Loss Function: 'squared_error' (optimizes for root mean squared error)

Here is the plot of the actual vs. predicted returns for the XGBoost implementation:

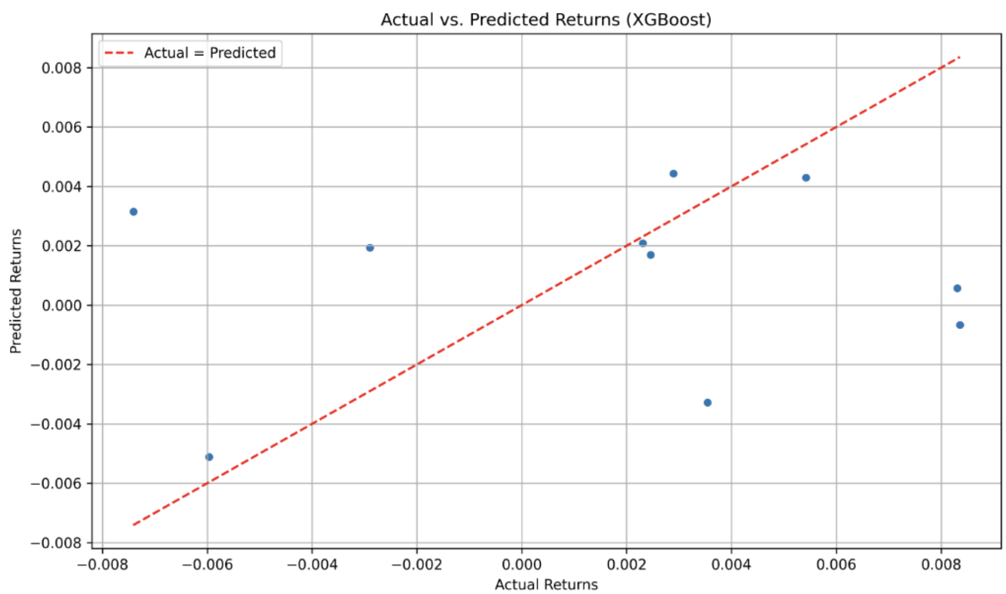

Figure 5: Illustrates the direct relationship between actual and predicted values

Additionally, here is the plot for the residuals against date_id:

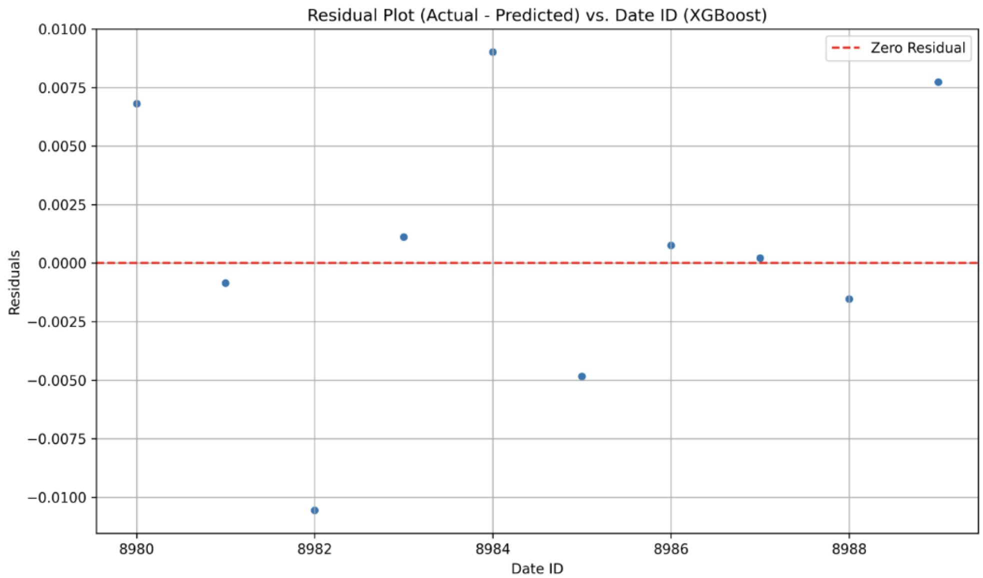

Figure 6: Residual plot

4 Discussion and Analysis

Our experiments with Gradient Boosting, XGBoost, and CatBoost reinforce a pattern that appears repeatedly in the literature on equity return prediction: it is possible to extract weak but non-trivial predictive signals from market data, but translating those signals into robust, risk-adjusted outperformance is difficult.

On one hand, our Gradient Boosting Regressor achieved a relatively high correlation (0.7124) between predicted and realized excess returns on the held-out test period. This suggests that the model is capturing directional structure in the data: periods when the model predicts above-average excess returns tend, on average, to coincide with higher realized returns. On the other hand, the strongly negative cross-validated R^2 value (-0.9520) points to poor generalization when the model is retrained on different folds of the historical data. In other words, the model can fit historical patterns extremely well in some splits but fails to explain much variance in others. This tension between a "promising signal" of the high correlation coefficient and "fragile generalization" of the R^2 is very representative of the current state of machine learning for market timing.

This same pattern is also seen with our CATBoost Regression model. Although, the model post a better time series cross-validated R^2 of -.5232, the model still fails to produce reliable generalized predictions. This is an extremely common issue among financial modeling data.

The improved cross-validated R^2 relative to the gradient boosting model is likely due to CATBoosting's handling of categorical data. Rather than utilizing some type of encoding such as one-hot encoding to transform categorical data, CATBoost is able to utilize the categorical data itself. The ordered target encoding that CATBoost utilizes diminishes the downfall of data leakage relative to other boosting models.

Although all three models are tree-based boosting methods, XGBoost likely underperformed due to a combination of data sensitivity, model complexity, and target characteristics in this financial forecasting setting. XGBoost is highly responsive to small shifts in feature distributions and requires careful hyperparameter tuning to avoid overfitting noisy, low-signal targets such as daily excess returns. In contrast, GradientBoostingRegressor and CatBoost are inherently more stable on noisy time-series data: Gradient Boosting uses shallow, conservative trees that smooth volatility, while CatBoost includes ordered boosting and strong built-in regularization that automatically mitigates noise, leakage, and small-sample instability.

Because the test set contains only 10 observations and the underlying target has an extremely low signal-to-noise ratio, models like the untuned XGBoost configuration that overfit even slightly, produce unstable predictions and weak directional correlation. Meanwhile, the more bias-leaning Gradient Boosting and CatBoost models generalize better under these conditions, leading to substantially higher test correlations compared to XGBoost.

One key takeaway is that high-capacity models are easy to overfit. Gradient boosting and its variants, as we have found, can aggressively chase spurious patterns in historical returns, especially when we introduce extensive feature engineering and imputation. Our cross-validation result suggests that some of the apparent signal may be partially driven by look-like noise that doesn't repeat in other periods. A positive correlation is encouraging, but a strongly negative R^2 underline that "explaining variance" in daily excess returns is extremely hard. Our work hints that purely predictive metrics are not sufficient, and that evaluation should be tied closely to a trading and risk-management framework.

From our results, we note a few directions that seem particularly promising:

- Shifting from point prediction to decision-focused modeling. Our model focused on minimizing generic regression loss and was evaluated via correlation and R^2. Recent work in "decision-aware" or "end-to-end" learning suggests directly optimizing for portfolio-level metrics (e.g., Sharpe-like objectives, downside risk penalties) rather than pure prediction error. In our setting, this would mean explicitly incorporating the 120% volatility constraint and our competition metric into training or model selection, rather than treating them as a post-hoc evaluation layer.
- Modeling uncertainty and distributions. Our Gradient Boosting Regressor produces point estimates of excess returns, implicitly assuming that only the conditional mean matters. In reality, tails and uncertainty are crucial: a slightly positive expected return with extremely high variance may be worse than a smaller but more stable signal. Methods such as quantile regression, Bayesian ensembles, or distributional models could provide predictive distributions over returns, enabling allocation rules that favor trades with attractive expected and risk characteristics.

Overall, our results support a nuanced view of market efficiency. The positive correlation suggests that markets may not be perfectly efficient, though there are some learnable patterns. However, the poor R^2 across folds emphasize that these patterns are weak, fleeting, and highly sensitive to modeling choices.

Our work with Gradient Boosting, XGBoost, and CatBoost forms a reasonable baseline in this landscape. It highlights both the existence of some predictive structure and the difficulty of turning that structure into stable, volatility-controlled outperformance.

5 References

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